10 Boolean logic

Candidates should be able to:	Notes and guidance
1 Identify and use the standard symbols for logic gates	• See section 4 for logic gate symbols
2 Define and understand the functions of logic gates	• Including: – NOT – AND – OR – NAND – NOR – XOR (EOR) – the binary output produced from all the possible binary inputs • NOT is a single input gate • All other gates are limited to two inputs
3 (a) Use logic gates to create given logic circuits from a: (i) problem statement (ii) logic expression (iii) truth table	• Circuits must be drawn for the statement given, without simplification • Logic circuits will be limited to a maximum of three inputs and one output
(b) Complete a truth table from a: (i) problem statement (ii) logic expression (iii) logic circuit	• An example truth table with three inputs, for completion:

A	B	C	Output
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(c) Write a logic expression from a: (i) problem statement (ii) logic circuit (iii) truth table	